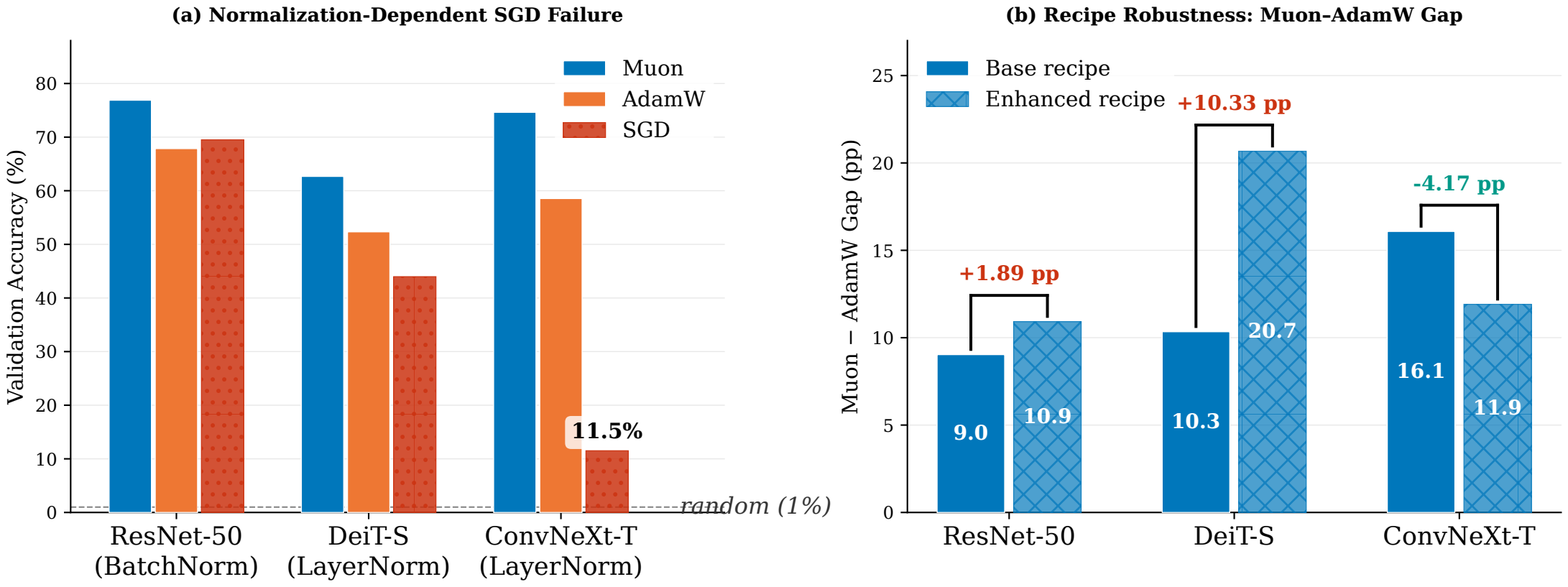
(a) Normalization-Dependent SGD Failure
Validation Accuracy (%)
Muon
AdamW
SGD
11.5%
random (1%)
ResNet-50 (BatchNorm)
DeiT-S (LayerNorm)
ConvNeXt-T (LayerNorm)
(b) Recipe Robustness: Muon–AdamW Gap
Muon − AdamW Gap (pp)
Base recipe
Enhanced recipe
+1.89 pp
+10.33 pp
-4.17 pp
9.0
10.9
10.3
20.7
16.1
11.9
ResNet-50
DeiT-S
ConvNeXt-T